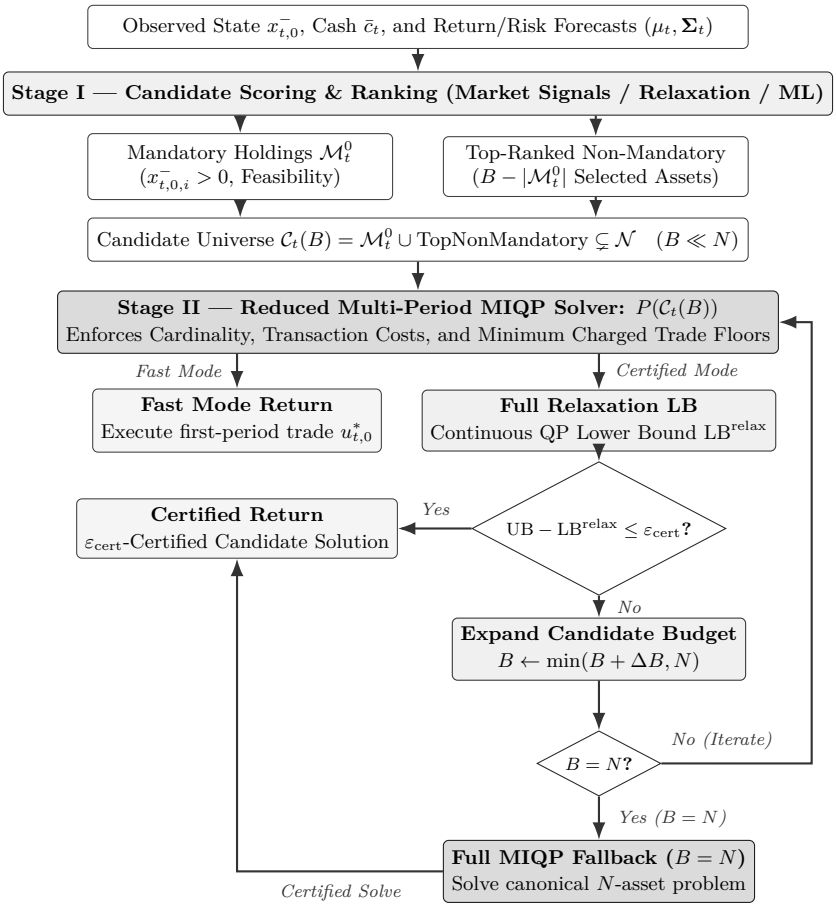
Observed State $x^{-}_{t,0}$, Cash $\bar{c}_t$, and Return/Risk Forecasts $(\mu_t, \mathbf{\Sigma}_t)$
Stage I — Candidate Scoring & Ranking (Market Signals / Relaxation / ML)
Mandatory Holdings $\mathcal{M}_t^0$ $(x^{-}_{t,0,i} > 0$, Feasibility)
Top-Ranked Non-Mandatory $(B - |\mathcal{M}_t^0|$ Selected Assets)
Candidate Universe $\mathcal{C}_t(B) = \mathcal{M}_t^0 \cup \text{TopNonMandatory} \subsetneq \mathcal{N} \quad (B \ll N)$
Stage II — Reduced Multi-Period MIQP Solver: $P(\mathcal{C}_t(B))$
Enforces Cardinality, Transaction Costs, and Minimum Charged Trade Floors
Fast Mode
Certified Mode
Fast Mode Return
Execute first-period trade $u^*_{t,0}$
Full Relaxation LB
Continuous QP Lower Bound LB^{relax}
Yes
$\text{UB} - \text{LB}^{\text{relax}} \leq \varepsilon_{\text{cert}}$?
Certified Return
$\varepsilon_{\text{cert}}$-Certified Candidate Solution
No
Expand Candidate Budget
$B \leftarrow \min(B + \Delta B, N)$
$B = N$?
No (Iterate)
Yes ($B = N$)
Full MIQP Fallback ($B = N$)
Solve canonical N-asset problem
Certified Solve